Day 3 Deliverable: Taxonomy, Resolver Groups and Ticket Schema

Workstream A (Data & Evaluation) | Version 0.2 | Aligned to the Workstream C data contract snapshot of 2026-10-01 | Draft for A/B/C/D review

---

1. What this is aligned to

- **Sources of truth:** the team's `taxonomy.json` (26 diagnosis codes), the 7 resolver group IDs, `routing_matrix.json`, `dataset.schemas.json` (ImportedTicket and others), `integration_contract.md`, and `escalation_policy.json` (`POLICY-DEFAULT`).
- **Rule:** nothing here renames a backend field, code, enum value or ID. Anything the backend does not support is labelled **Proposal** and needs A/B/C/D agreement before it is built.
- **Checked against the per-record schemas you provided.** It has not been run through the real importer. Run `python -m scripts.import_data --data-dir <dir> --validate-only` from the backend root once the full file set exists (it needs every operational file, not only these).

File	Purpose
`data/taxonomy.json`	Exact import shape: flat array, 7 schema fields per record, same 26 codes as the team file
`data/resolver_groups.json`	Exact import shape: `{id, label}` for the 7 groups
`data/generation_plan.json`	A's planning data (volumes, fixtures, clusters, decoys). **Not an importer input**
`data/build_day3.py`	Regenerates all of the above and self-checks the examples in section 5

---

2. Taxonomy

Three levels: **Domain > Category > Diagnosis code**. The domain uses the four reporting buckets from the project brief (exact strings, no aliases). 20 categories and 26 diagnosis codes. The `diagnosis_code` is the machine key; the `label` is what the backend writes into `ticket.subcategory`.

2.1 Domain targets (500 historical tickets)

Domain	Target share	Historical tickets	Categories	Diagnosis codes
Access Management	40%	200	2	6
Application	35%	175	5	5
Endpoint	10%	50	3	3
Network/Other	15%	75	10	12

2.2 Codes by domain

Access Management (~40%)

Diagnosis code	Category	Label (= ticket.subcategory)	Owner group	Security	Privileged	Hist. tickets
`PASSWORD_EXPIRED`	Authentication	Expired password	`GRP-IAM`	false	false	55
`ACCOUNT_LOCKED`	Authentication	Account lockout	`GRP-IAM`	false	false	45
`ACCOUNT_LOCKED_POLICY`	Authentication	Account locked by security policy	`GRP-SEC`	true	false	15
`ACCOUNT_DISABLED`	Authentication	Disabled account	`GRP-IAM`	false	false	20

Diagnosis code	Category	Label (= ticket.sub category)	Owner group	Security	Privileged	Hist. tickets
`MFA_FAILURE`	Authentication	MFA failure	`GRP-IAM`	false	false	50
`PRIVILEGED_ACCESS`	Privileges	Privileged-access request	`GRP-IAM`	false	true	15

Application (~35%)

Diagnosis code	Category	Label (= ticket.sub category)	Owner group	Security	Privileged	Hist. tickets
`MISSING_ENTITLEMENT`	Authorization	Missing application entitlement	`GRP-APP`	false	false	50
`APP_AUTH_FAILURE`	Authentication	Application authentication failure	`GRP-APP`	false	false	40
`APP_OUTAGE`	Availability	Application service outage	`GRP-APP`	false	false	25
`APP_ERROR`	Execution	Application execution error	`GRP-APP`	false	false	40
`SYNC_DELAY`	Synchronization	Application synchronization delay	`GRP-APP`	false	false	20

Endpoint (~10%)

Diagnosis code	Category	Label (= ticket.sub category)	Owner group	Security	Privileged	Hist. tickets
LOCAL_ENDPOINT	Device	Local endpoint issue	GRP-END	false	false	15
DISK_FULL	Storage	Endpoint disk full	GRP-END	false	false	17
PRINTER_OFFLINE	Printing	Printer offline	GRP-END	false	false	18

Network/Other (~15%)

Diagnosis code	Category	Label (= ticket.sub category)	Owner group	Security	Privileged	Hist. tickets
VPN_OUTAGE	VPN	VPN outage	GRP-NET	false	false	10
VPN_CLIENT_ISSUE	VPN	VPN client or credential issue	GRP-NET	false	false	15
DNS_FAILURE	DNS	DNS lookup failure	GRP-NET	false	false	6
WIFI_FAILURE	Connectivity	Wi-Fi association failure	GRP-NET	false	false	8
PHISHING	Threat	Suspected phishing	GRP-SEC	true	false	8
MALWARE	Threat	Suspected malware	GRP-SEC	true	false	5

Diagnosis code	Category	Label (= ticket.sub category)	Owner group	Security	Privileged	Hist. tickets
DATA_EXPOSURE	Data Protection	Suspected data exposure	GRP-SEC	true	false	3
SERVER_FAILURE	Compute	Server failure	GRP-INFRA	false	false	5
STORAGE_FAILURE	Storage	Storage service failure	GRP-INFRA	false	false	3
CERT_EXPIRED	Certificates	Expired service certificate	GRP-INFRA	false	false	4
UNKNOWN_ISSUE	Unclassified	Insufficient evidence	GRP-L1	false	false	4
REQUEST_GUIDANCE	How To	General guidance request	GRP-L1	false	false	4

2.3 Notes

- Codes, domains, categories, labels, groups and flags are copied unchanged from the team file. **Only the per-code historical volumes are new** (A's planning data).
- The team file also carries `sub_domain`, `classification_path`, `human_required` and `outage_capable`. The per-record schema forbids extra fields, so `taxonomy.json` here omits them. `sub_domain` (the old seven-domain view) and `outage_capable` are kept in `generation_plan.json`; `classification_path` and `human_required` are derivable.
- The category **Authentication** exists under both Access Management and Application. Counts and searches by category alone mix the two; add the domain filter.
- No new codes were added. Any new code needs a routing-matrix row, an article from B, and agreement with C.

---

3. Resolver groups

Exactly the 7 backend groups. Use the **ID**, never the display name.

ID	Label	Codes	Diagnoses owned (from taxonomy)
GRP-L1	Service Desk L1	2	UNKNOWN_ISSUE, REQUEST_GUIDANCE
GRP-IAM	IAM	5	PASSWORD_EXPIRED, ACCOUNT_LOCKED, ACCOUNT_DISABLED, MFA_FAILURE, PRIVILEGED_ACCESS
GRP-APP	Application Support	5	MISSING_ENTITLEMENT, APP_AUTH_FAILURE, APP_OUTAGE, APP_ERROR, SYNC_DELAY
GRP-END	Endpoint Support	3	LOCAL_ENDPOINT, DISK_FULL, PRINTER_OFFLINE
GRP-NET	Network Operations	4	VPN_OUTAGE, VPN_CLIENT_ISSUE, DNS_FAILURE, WIFI_FAILURE
GRP-SEC	Security Operations	4	ACCOUNT_LOCKED_POLICY, PHISHING, MALWARE, DATA_EXPOSURE
GRP-INFRA	Servers & Infrastructure	3	SERVER_FAILURE, STORAGE_FAILURE, CERT_EXPIRED

3.1 Boundary rules (derived from the taxonomy and routing matrix)

The routing matrix states one shared boundary: *IAM owns identity; Application Support owns app-specific sessions and entitlements; Security leads suspected threats; L1 triages uncertainty.* In practice:

Situation	**Rule**
Password expired, lockout, disabled account, MFA	`GRP-IAM`. If the account was locked by a security policy: `ACCOUNT_LOCKED_POLICY` goes to `GRP-SEC`, always with a human.
Cannot access an application	Identity problem (password, lock, MFA): `GRP-IAM`. Missing entitlement or application-side authentication failure: `GRP-APP`. Application down for many users: `APP_OUTAGE`, `GRP-APP`, plus the outage path.
VPN not connecting	`VPN_CLIENT_ISSUE` (client **or credential**) and `VPN_OUTAGE` both go to `GRP-NET`.
Slow or failing application	Application behaviour (`APP_ERROR`, `SYNC_DELAY`): `GRP-APP`. Server or storage failure: `GRP-INFRA`. Device-side problem: `GRP-END`.
Privileged or admin access	`GRP-IAM` provisions, but `privileged_access=true` always requires a human. Review approval never allows agent resolution.
Phishing, malware, data exposure	`GRP-SEC`, always human.
Expired service certificate	`GRP-INFRA`.
Vague, unclassifiable or how-to	`GRP-L1`. The agent asks questions and must not force a diagnosis.

3.2 Routing outcome by confidence (from the integration contract)

Confidence	Backend behaviour
HIGH	Assigned to the requested specialist group, subject to an approved human review
MEDIUM	Assigned to `GRP-L1`; the specialist recommendation and up to 3 alternatives are kept
LOW	`GRP-L1`, status `PENDING`, human triage required

Ground truth stores one correct group. **Top-3 accuracy** compares it with the agent's ranked list of up to three groups.

---

4. Escalation policy and what it means for the data

`POLICY-DEFAULT` as provided (B owns the final values):

Item	Value
Mandatory human cases	`security_sensitive`, `privileged_access`, `reported_priority=1`
SEV1	Critical business impact; human review required
SEV2	Multiple affected users in one service scope
SEV3	Limited service degradation

Category	Configuration item	Location	Window (min)	Minimum count
VPN	`CI-VPN`	`LOC-A`	30	3
Availability	`CI-PORTAL`	`LOC-B`	30	3

Category	Configuration item	Location	Window (min)	Minimum count
Availability	CI-ERP	LOC-C	30	3

Consequences for Workstream A:

- **Derived labels.** `human_required` = taxonomy `security_sensitive` or `privileged_access`, or `reported_priority=1`. A P1 `PASSWORD_EXPIRED` ticket is therefore human-required and **not** agent-resolvable.
- **Outage scopes are exact.** A cluster only counts if every ticket shares the same category, configuration item and location, and these must match one threshold row. Only VPN and Availability have thresholds; `SERVER_FAILURE` and `STORAGE_FAILURE` are outage-capable but cannot form a cluster under this policy.
- **The starter threshold is 3 tickets in 30 minutes**, not the brief's illustrative 20. Cluster size is a generator parameter (default 8).

---

5. Ticket schema

5.1 Three ticket shapes (do not mix them)

Shape	Who	Notes
CreateTicket request	Agent D, `POST /tickets`	Only `action_id`, `summary`, `description`, `reported_by_id` required. Backend creates ID, status, timestamps, first worklog
ImportedTicket record	Workstream A, dataset importer	Flat record, section 5.2. No nested worklogs or evidence. Extra fields are rejected
TicketDetails response	`GET /tickets/{id}`	Adds `related_ticket_ids`, `worklogs`, `evidence`, `latest_diagnostic_result`. Never send it back to the importer

A nullable field uses JSON `null`, never `""`, `"N/A"` or `"unknown"`. Timestamps must carry a timezone and are normalised to UTC (`2026-10-01T12:00:00Z`).

5.2 ImportedTicket fields with the Maximo IT mapping

Maximo attribute names are from the standard TICKET objects and **should be verified against the Maximo IT documentation**. 'custom' means no direct Maximo field.

Field	Type / rule	Notes	Maximo field
`ticket_id`	string, required	Stable ID. Prefixes are convention only	TICKETID
`ticket_class`	enum, required	SERVICE_REQUEST, INCIDENT, PROBLEM, CHANGE	CLASS
`summary`	string, required	Never empty	DESCRIPTION
`description`	string, required	Unicode: English, Arabic, Franco-Arabic. Never empty	DESCRIPTION_LONGDESCRIPTION
`status`	enum, required	NEW, QUEUED, IN_PROGRESS, PENDING, RESOLVED, CLOSED	STATUS
`reported_priority`	int 1-4, required	1 = critical (forces human handling)	REPORTEDPRIORITY
`impact`	int 1-3, required	1 = highest	IMPACT
`urgency`	int 1-3, required	1 = highest	URGENCY
`language`	enum, required	en, ar, franco_ar only	custom
`domain`	string or null	Historical: taxonomy domain. Individual test tickets: null	classification hierarchy

Field	Type / rule	Notes	Maximo field
`category`	string or null	As above	classification hierarchy
`subcategory`	string or null	Historical: taxonomy label	classification hierarchy
`diagnosis_code`	string or null	Historical: required. Open: must be null	custom
`owner_group_id`	string or null	`GRP-*`. Historical: final group. Open: normally null	OWNERGROUP
`owner_id`	string or null	Normally null	OWNER
`reported_by_id`	string, required	ID from `users.json`	REPORTEDBY
`affected_person_id`	string, required	ID from `users.json`	AFFECTEDPERSON
`asset_id`	string or null	ID from `assets.json`	ASSETNUM
`configuration_item_id`	string or null	ID from `configuration_items.json`	CINUM
`location_id`	string or null	ID from `locations.json`	LOCATION
`reported_at`	datetime, required	UTC, not after the demo clock	REPORTDATE
`target_finish_at`	datetime or null	Derived from priority	TARGETFINISH
`actual_finish_at`	datetime or null	Historical only	ACTUALFINISH
`resolved_at`	datetime or null	Historical: required	status history (derived)
`closed_at`	datetime or null	Required when CLOSED	status history (derived)
`updated_at`	datetime, required	Last change	CHANGEDATE

Field	Type / rule	Notes	Maximo field
`linked_solution_id`	string or null	Open: must be null. Historical: B's approved article ID, or null	linked SOLUTION
`resolution_summary`	string or null	Open: must be null. Historical: required	resolution text
`is_global`	boolean	Must be false on import	GLOBALISSUE
`global_ticket_id`	string or null	Must be null on import	global-issue relationship
`severity`	SEV1/SEV2/SEV3 or null	Null on import (outage tickets only)	custom
`security_sensitive`	boolean	Safety flag, default false	custom
`privileged_access`	boolean	Safety flag, default false	custom
`initial_owner_group_id`	string or null	**Import-only.** First (wrong) group of a misrouted historical ticket	first owner-history entry

Response-only fields (never importable): `related_ticket_ids`, `worklogs`, `evidence`, `latest_diagnostic_result`.

5.3 Enums A uses

Concept	Accepted values
ticket_class	SERVICE_REQUEST, INCIDENT, PROBLEM, CHANGE
status	NEW, QUEUED, IN_PROGRESS, PENDING, RESOLVED, CLOSED
language	en, ar, franco_ar

Concept	Accepted values
severity	SEV1, SEV2, SEV3 (null otherwise)
account / entitlement state	**strings** `TRUE`, `FALSE`, `UNKNOWN`
service health state	OPERATIONAL, DEGRADED, UNAVAILABLE, UNKNOWN
provisioning state	ACTIVE, PENDING, ABSENT, UNKNOWN
safety flags and `agent_resolvable`	ordinary JSON booleans `true` / `false`

5.4 Rules by ticket kind

Kind	Rules
Historical (400-600)	RESOLVED or CLOSED. Needs `diagnosis_code`, `resolution_summary`, `resolved_at` (and `closed_at` if CLOSED). Classification fields equal the taxonomy. Misrouting only via `initial_owner_group_id` (single hop, about 15%)
Individual test ticket (80-100)	Status NEW. No diagnosis, solution or resolution. `domain`, `category`, `subcategory` null. Missing optional fields = null `asset_id`, `configuration_item_id` or `location_id`
Cluster ticket (outage or decoy)	As a test ticket, but **`category`, `configuration_item_id` and `location_id` must be filled**, because the count tool matches stored ticket fields. `diagnosis_code` stays null
Neutral IDs	Use `HIS-0001...` for historical and `TST-0001...` for every test ticket, cluster tickets included. IDs like `CLUSTER-1-1` would leak the answer to the agent; cluster membership lives only in ground truth

5.5 Worklogs and evidence: not importable today

The importer generates an initial log and, if `initial_owner_group_id` is set, the reassignment log. It **cannot** accept A's authored troubleshooting worklogs or evidence, and closed tickets accept no new writes. A and C must agree an importer extension first. **Proposal** (shapes match the response objects):

File	Record fields
`historical_worklogs.json`	`worklog_id, ticket_id, timestamp, actor_id, entry_type, summary, rationale, knowledge_article_ids, historical_ticket_ids, evidence_ids, action_id (null)`. Types: NOTE, QUESTION, EVIDENCE, DIAGNOSIS, ASSIGNMENT, RESOLUTION
`historical_evidence.json`	`evidence_id, ticket_id, source_type, source_id, observation, observed_at`. For USER_RESPONSE, `source_id` = the affected person

5.6 Hidden evaluation files (Proposal)

Stored outside the runtime import, under `evaluation/` (not in the database import or Docker image), joined to tickets by `ticket_id`. I have **not seen `examples.json`**, so these field names must be reconciled with its `proposed_evaluation_record`. The starter file uses `expected_diagnosis`; this proposal uses `expected_diagnosis_code`. A and D pick one before any evaluator is written.

Field	Meaning / how it is filled
`ticket_id`, `evaluation_only`	Join key and marker (as in the starter file)
`scenario_type`	normal, ambiguous, security, vague, outage, decoy
`ambiguity_cluster`, `cluster_id`	AC-n or null; OUT-n / DECOY-n or null
`expected_domain`, `expected_category`	**Derived** from the taxonomy code
`expected_diagnosis_code`	Authored label

Field	Meaning / how it is filled
`expected_resolver_group_id`	**Derived** from the taxonomy code
`agent_resolvable`	**Derived**: approved article is agent-resolvable and no human-required condition applies
`human_required`	**Derived**: security_sensitive or privileged_access or reported_priority=1
`escalation_required`	**Derived**: human_required, or the ticket belongs to an outage cluster
`initial_candidates`, `initial_information`	Plausible diagnoses and facts visible at the start
`discoverable_information`	List of `{topic, answer, eliminates}`: scripted answers, in the ticket's language, that the agent only gets by asking
`required_fixtures`	Account, entitlement or health values that must exist in the fixture files so tool checks agree with the truth

5.7 Examples (self-checked against the ImportedTicket schema)

Illustrative IDs such as `USR-014` must exist in the reference files at import time.

Historical ticket, misrouted to IAM first, then correctly owned by Application Support (the importer writes the reassignment log):

{

"ticket_id": "HIS-0042",

"ticket_class": "INCIDENT",

"summary": "Cannot open ERP, says no access",

"description": "Since this morning ERP shows an access denied page. Email and the portal work normally.",

"status": "CLOSED",

"reported_priority": 3,

"impact": 3,

"urgency": 2,

"language": "en",

"domain": "Application",

"category": "Authorization",

"subcategory": "Missing application entitlement",

"diagnosis_code": "MISSING_ENTITLEMENT",

"owner_group_id": "GRP-APP",

"owner_id": null,

"reported_by_id": "USR-014",

"affected_person_id": "USR-014",

"asset_id": "ASSET-014",

"configuration_item_id": "CI-ERP",

"location_id": "LOC-C",

"reported_at": "2026-06-02T08:14:00Z",

"target_finish_at": "2026-06-04T08:14:00Z",

"actual_finish_at": "2026-06-02T13:40:00Z",

"resolved_at": "2026-06-02T13:40:00Z",

"closed_at": "2026-06-04T08:00:00Z",

"updated_at": "2026-06-04T08:00:00Z",

"linked_solution_id": null,

"resolution_summary": "ERP user role was missing. Role added after manager approval and the user confirmed access.",

"is_global": false,

"global_ticket_id": null,

"severity": null,

"security_sensitive": false,

"privileged_access": false,

"initial_owner_group_id": "GRP-IAM"

}

Individual test ticket (visible part only; Franco-Arabic, ambiguous cluster AC-1):

{

"ticket_id": "TST-0017",

"ticket_class": "INCIDENT",

"summary": "el application msh shaghal",

"description": "el application msh byefta7 w ana mesh 3arf leh",

"status": "NEW",

"reported_priority": 3,

"impact": 3,

"urgency": 2,

"language": "franco_ar",

"domain": null,

"category": null,

"subcategory": null,

"diagnosis_code": null,

"owner_group_id": null,

"owner_id": null,

"reported_by_id": "USR-073",

"affected_person_id": "USR-073",

"asset_id": null,

"configuration_item_id": null,

"location_id": "LOC-B",

"reported_at": "2026-10-01T09:30:00Z",

"target_finish_at": null,

"actual_finish_at": null,

"resolved_at": null,

"closed_at": null,

"updated_at": "2026-10-01T09:30:00Z",

"linked_solution_id": null,

"resolution_summary": null,

"is_global": false,

"global_ticket_id": null,

"severity": null,

"security_sensitive": false,

```
  "privileged_access": false,
  "initial_owner_group_id": null
}
```


Its hidden ground truth (`evaluation/`):


```
{
  "ticket_id": "TST-0017",
  "evaluation_only": true,
  "scenario_type": "ambiguous",
  "ambiguity_cluster": "AC-1",
  "cluster_id": null,
  "expected_domain": "Application",
  "expected_category": "Authorization",
  "expected_diagnosis_code": "MISSING_ENTITLEMENT",
  "expected_resolver_group_id": "GRP-APP",
  "agent_resolvable": false,
  "human_required": false,
  "escalation_required": false,
  "initial_candidates": [
    "PASSWORD_EXPIRED",
    "ACCOUNT_LOCKED",
    "ACCOUNT_DISABLED",
    "MFA_FAILURE",
```

```
    "MISSING_ENTITLEMENT",
    "APP_AUTH_FAILURE",
    "APP_OUTAGE"
  ],
  "initial_information": [
    "Franco-Arabic text",
    "an application does not open",
    "no application named",
    "no error message"
  ],
  "discoverable_information": [
    {
      "topic": "other_corporate_applications",
      "answer": "el email we el HR system sheghalin tamam, el portal bas el msh byefta7",
      "eliminates": [
        "PASSWORD_EXPIRED",
        "ACCOUNT_LOCKED",
        "ACCOUNT_DISABLED",
        "MFA_FAILURE"
      ]
    },
```

```
  {
    "topic": "which_application",
    "answer": "el service portal",
    "eliminates": []
  },
  {
    "topic": "error_message",
    "answer": "bey2olly 'You do not have access'",
    "eliminates": [
      "APP_AUTH_FAILURE",
      "APP_OUTAGE"
    ]
  }
],
"required_fixtures": {
  "check_account_status": {
    "enabled": "TRUE",
    "locked": "FALSE",
    "password_expired": "FALSE",
    "mfa_enrolled": "TRUE",
    "mfa_locked": "FALSE"
  },
```

```
    "check_application_entitlement": {
      "application_id": "APP-PORTAL",
      "entitled": "FALSE",
      "provisioning_state": "ABSENT"
    }
  }
}
```


---

6. Expected tool-check facts per diagnosis (Proposal)

Tool checks read static fixtures, so the fixture rows for each test ticket's user, CI and location must be consistent with its hidden diagnosis. Codes not listed have no supporting tool check; their evidence comes from the user's answers.

Diagnosis code	check_account_status	check_application_entitlement	check_service_health
`PASSWORD_EXPIRED`	enabled=TRUE, locked=FALSE, password_expired=TRUE	-	-
`ACCOUNT_LOCKED`	enabled=TRUE, locked=TRUE, password_expired=FALSE	-	-
`ACCOUNT_LOCKED_POLICY`	enabled=TRUE, locked=TRUE, password_expired=FALSE	-	-
`ACCOUNT_DISABLED`	enabled=FALSE	-	-

Diagnosis code	check_account_status	check_application_entitlement	check_service_health
MFA_FAILURE	enabled=TRUE, locked=FALSE, password_expired=FALSE, mfa_locked=TRUE **or** enabled=TRUE, locked=FALSE, password_expired=FALSE, mfa_enrolled=FALSE	-	-
PRIVILEGED_ACCESS	enabled=TRUE, locked=FALSE, password_expired=FALSE	-	-
MISSING_ENTITLEMENT	enabled=TRUE, locked=FALSE, password_expired=FALSE	entitled=FALSE, provisioning_state=ABSENT **or** entitled=FALSE, provisioning_state=PENDING	-
APP_AUTH_FAILURE	enabled=TRUE, locked=FALSE, password_expired=FALSE	entitled=TRUE, provisioning_state=ACTIVE	state=OPERATIONAL
APP_OUTAGE	-	-	state=UNAVAILABLE **or** state=DEGRADED
APP_ERROR	enabled=TRUE, locked=FALSE, password_expired=FALSE	entitled=TRUE, provisioning_state=ACTIVE	state=OPERATIONAL
SYNC_DELAY	enabled=TRUE, locked=FALSE, password_expired=FALSE	-	state=OPERATIONAL **or** state=DEGRADED

Diagnosis code	check_account_status	check_application_entitlement	check_service_health
`VPN_OUTAGE`	-	-	state=UNAVAILABLE **or** state=DEGRADED
`VPN_CLIENT_ISSUE`	enabled=TRUE, locked=FALSE, password_expired=FALSE	-	state=OPERATIONAL
`SERVER_FAILURE`	-	-	state=UNAVAILABLE
`STORAGE_FAILURE`	-	-	state=UNAVAILABLE
`CERT_EXPIRED`	-	-	state=DEGRADED **or** state=UNAVAILABLE

Important limit: `ACCOUNT_LOCKED` and `ACCOUNT_LOCKED_POLICY` look identical to the account check (`locked=TRUE`). Only the user's answers and the security flag tell them apart. That is a deliberate hard case for the test set.

---

7. Synthetic data rules

Rule	Value
Random seed	`42`
Demo clock	`2026-10-01T12:00:00Z`. Historical `reported_at` between 2026-04-01 and 2026-09-24. Individual test tickets within the 3 days before the clock. Cluster tickets relative to the clock
IDs	`HIS-`, `TST-` for tickets; reference IDs use `USR-`, `ASSET-`, `CI-`, `LOC-`, `GRP-`; keep the starter IDs (`USR-001`, `ASSET-001`, `CI-PORTAL`, `CI-VPN`, `CI-ERP`, `LOC-A`..`LOC-C`, `APP-PORTAL`)

Rule	Value
Reference data	Proposal: A drafts `users.json`, `assets.json`, `configuration_items.json`, `locations.json` (LOC-A..LOC-F) and the account, entitlement and health fixtures; C approves
Language mix	English 50%, Arabic 30%, Franco-Arabic 20%
Priority mix	P1 5%, P2 20%, P3 50%, P4 25%. Impact and urgency use 1-3
Noise	Typos ~20%, vague ~8%, missing optional fields ~15%. `summary` and `description` are never empty
Misrouting	~15% of historical tickets (75 of 500) via `initial_owner_group_id`. Plausible wrong groups per code are in `generation_plan.json`
Fake sensitive content	~8% of tickets. Use `SYNTH_SECRET_nnn` markers (the format the backend redacts) and IPs from `192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24`. Emails only `@corp.example`. No real names, phones or client name
Test set	90 individual tickets (36 normal, 28 ambiguous = 4 per cluster, 14 security or human-required, 12 vague) plus 6 cluster scenarios

---

8. Outage and decoy design

Health fixtures are **static** per configuration item and location, so a decoy must **not** reuse an outage scope (its health would read UNAVAILABLE and wrongly signal an outage).

Outage clusters (one per policy threshold; all tickets within 30 minutes of the clock; expected: global ticket created, later tickets linked, owner group notified):

ID	Category	CI	Location	Tickets	Health fixture	Global owner	Expected severity
OUT-1	VPN	CI-VPN	LOC-A	8	UNAVAILABLE	GRP-NET	SEV2
OUT-2	Availability	CI-PORTAL	LOC-B	8	UNAVAILABLE	GRP-APP	SEV2
OUT-3	Availability	CI-ERP	LOC-C	8	DEGRADED	GRP-APP	SEV2

Replay tickets in time order with `as_of` = the ticket's `reported_at`: the first two are diagnosed normally, the third meets the threshold.

Decoy clusters (must **not** trigger escalation; health fixtures OPERATIONAL):

ID	Pattern	Category	CI	Locations	Shape
DECOY-1	same scope, spread over days	VPN	CI-VPN	LOC-D	8 tickets over 5 days
DECOY-2	same time, spread over sites	Availability	CI-PORTAL	LOC-A, LOC-C, LOC-D	7 tickets inside one 30-minute window, at most 2 per site
DECOY-3	spread over days and sites	Availability	CI-ERP	LOC-D, LOC-E	6 tickets over 3 days, at most 2 per site per 30 minutes

A decoy only proves something if a threshold exists for its scope. **Request to B:** add these 6 threshold rows (same 3 tickets in 30 minutes):

Category	CI	Location
VPN	`CI-VPN`	`LOC-D`
Availability	`CI-PORTAL`	`LOC-A`
Availability	`CI-PORTAL`	`LOC-C`
Availability	`CI-PORTAL`	`LOC-D`
Availability	`CI-ERP`	`LOC-D`
Availability	`CI-ERP`	`LOC-E`

---

9. Open items

#	Item	Owners
1	Guide text says 24 codes/articles; the taxonomy file has 26. `routing_matrix.json` has 24 rows and no row for `ACCOUNT_LOCKED_POLICY` or `VPN_CLIENT_ISSUE`	B, C
2	Guide text lists 7 domains; the taxonomy file uses the 4 brief names as `domain` (old 7 as `sub_domain`). This pack follows the file	A, B, C
3	Team file has extra taxonomy fields that the per-record schema forbids. Run `--validate-only` to see which is right	C
4	`examples.json` not seen; evaluation field names in 5.6 are proposals (`expected_diagnosis` vs	A, D

#	Item	Owners
	`expected_diagnosis_code`)	
5	Reference data and fixtures must exist for every ID A uses; ownership and extra IDs to be agreed	A, C
6	Importer extension for historical worklogs and evidence (5.5)	A, C
7	Final outage thresholds; 6 extra rows for decoys; cluster size is a parameter	B, A
8	Test tickets arrive with `security_sensitive` and `privileged_access` false, so the agent must detect them; only `reported_priority=1` is known at intake	A, D
9	May ticket descriptions contain `SYNTH_SECRET_` markers? The contract only mentions notes and details	C
10	Only KB-001 is agent-resolvable today; `agent_resolvable` in ground truth is derived from approved articles at build time	B, A
11	Do 90 individual tickets plus 6 cluster scenarios satisfy the '80-100' target?	Karim
12	Verify Maximo attribute names in the mapping	A

#	Item	Owners
13	Name this snapshot `contract-v1` only after the owners agree; keep a contract change log	All

---

10. Next steps (Day 4)

1. Review this pack with B, C and D; settle items 1-8.
2. Draft the reference files and fixtures (A), approved by C.
3. Write `generate_data.py` (seed 42) that reads `taxonomy.json` and `generation_plan.json` and enforces the validation rules in `generation_plan.json`.
4. Generate the 500 historical tickets, then validate with the importer.